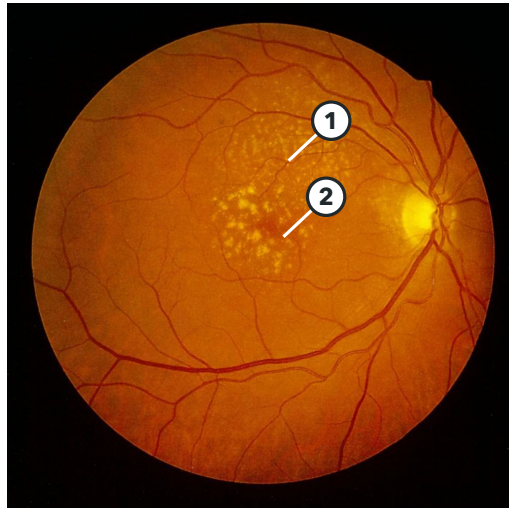


Macular and vitreoretinal disease

Clinical M3-M5 | Central distortion, flashes, floaters and a curtain each lead to a different pathway.

Macular disease: read the centre of the fundus

Drusen at the macula



1: yellow drusen clustered around the macula. 2: macular region. This photo illustrates deposits; it does not by itself establish active neovascular AMD. Supplied M3 Anki, media 168.

Localise the visual complaint

Complaint	Likely site / next step
Wavy lines or central scotoma	Macula; check each eye and obtain macular OCT when indicated
Peripheral curtain or shadow	Possible detachment; urgent dilated retinal assessment
New flashes and mobile floaters	Vitreoretinal traction/PVD; exclude a peripheral retinal tear

Dry and wet are clinical pathways

Drusen/RPE change can progress to atrophy. New macular fluid or haemorrhage suggests neovascular disease. Compare acuity, Amsler distortion, the fundus and OCT rather than diagnosing from a yellow spot alone.

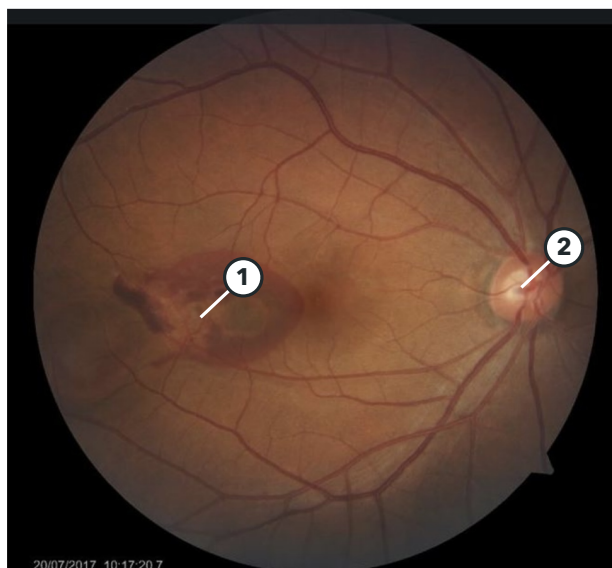
Prevention and monitoring

Encourage smoking cessation and symptom self-monitoring. AREDS2 is for appropriate intermediate AMD or late AMD in one eye, not a universal supplement for every patient with early drusen.

References NICE NG82: Age-related macular degeneration | NEI: AREDS2 supplements and AMD stage | NEI: AREDS and AREDS2 trials

Suspected wet AMD: prompt macular assessment

Haemorrhage centred on the macula



1: subretinal/sub-RPE haemorrhage at the macula. 2: optic disc for orientation. In an older adult with new distortion, this pattern warrants an urgent neovascular AMD assessment. Supplied M3 Anki, media 299.

Investigate and treat the active lesion

- 1 New metamorphopsia or central blur: refer urgently to a macula service.
- 2 Macular OCT assesses intra/subretinal fluid and structural change; add fluorescein angiography if neovascular disease remains uncertain.
- 3 Confirmed active neovascular AMD: retinal team plans intravitreal anti-VEGF and response monitoring.

Distinguish the endpoint

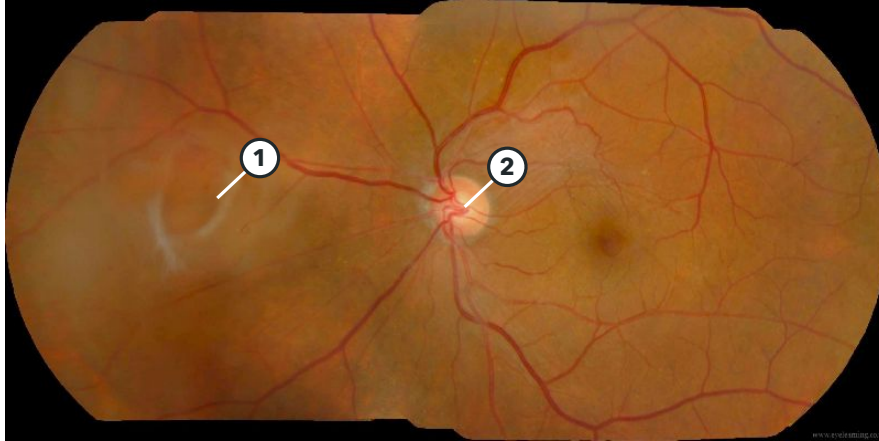
Atrophy or an established scar may cause central loss without active leakage. Treatment decisions depend on activity and expected benefit, not simply on the label “AMD”.

Useful bedside test

An Amsler grid demonstrates central distortion when each eye is tested separately. It supports symptom monitoring but cannot replace retinal examination or OCT.

Acute PVD: the Weiss ring is only part of the assessment

Posterior hyaloid separation



1: Weiss ring, an opacity released from the disc attachment. 2: disc. A ring identifies PVD; it does not show that the peripheral retina is free of tears. Supplied M3 Anki, media 181.

Before reassurance

- 1 Ask about onset, flashes, a shower of floaters, reduced vision and a peripheral shadow.
- 2 Measure acuity; inspect anterior vitreous for pigment/haemorrhage and examine the entire peripheral retina with appropriate dilation/indentation.
- 3 Tear or detachment: urgent retinal treatment. Uncomplicated PVD: safety-net and arrange follow-up according to risk.

Features that raise the stakes

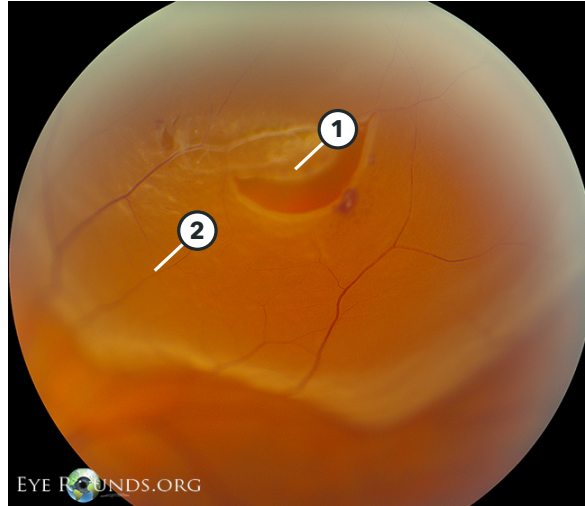
Finding	Why it matters
Vitreous pigment or fresh haemorrhage	A retinal break is more likely
New field shadow or falling acuity	Possible detachment or associated bleed
High myopia, trauma or prior eye surgery	Greater risk of retinal breaks

Safety-net explicitly

Return immediately for increased floaters/flashes, a curtain or reduced vision, even after an initially negative examination. New tears can develop during evolving PVD.

Retinal detachment: protect the attached macula

A break with surrounding detached retina



1: horseshoe-shaped retinal tear. 2: elevated grey retina with altered surface contour. This peripheral image cannot determine whether the macula remains attached. Supplied M3 Anki, media 161.

Three mechanisms, three routes

Type	Typical clue / treatment direction
Rhegmatogenous	Retinal break lets fluid underneath; urgent vitreoretinal repair planning
Tractional	Fibrovascular tissue pulls retina, often in PDR; assess macular threat and surgery
Exudative	Fluid without a primary break; identify inflammation, tumour or another cause

Urgency is not decided by acuity alone

New flashes/floaters plus a spreading curtain require same-day ophthalmic assessment. A macula-on detachment can retain central acuity while threatening it; the retina team decides repair timing.

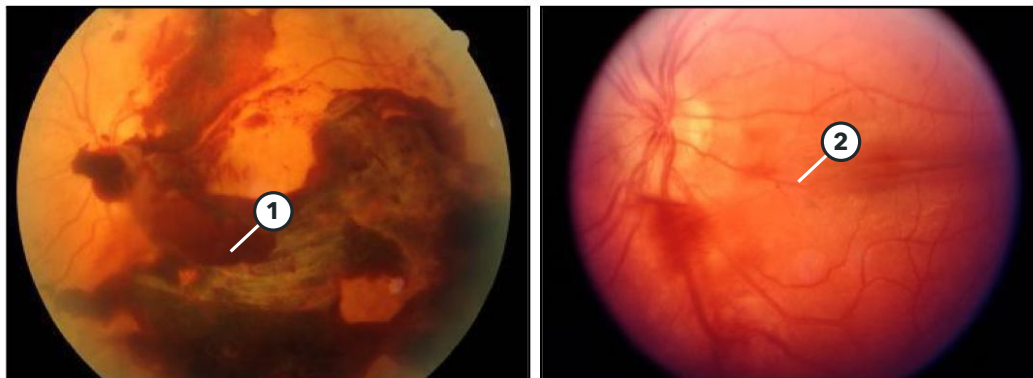
Examine beyond the posterior pole

Map the detachment and breaks, assess macular status, and examine the fellow eye. B-scan helps when media opacity prevents a reliable retinal view; a central photograph cannot exclude a peripheral tear.

References NEI: Retinal detachment and urgent assessment | AAO EyeWiki: Retinal detachment examination | AAO: PVD and Retinal Breaks PPP (2025)

Vitreous haemorrhage: find the cause behind the blood

A bleed can obscure the retinal examination



1: dense vitreous blood obscures the fundus in the left panel. 2: a clearer fundus view in the right panel. Media opacity does not remove the need to exclude a break or detachment. Supplied M3 Anki, media 221.

Investigate an obscured fundus

- 1** Sudden painless floaters/haze or loss of vision: assess acuity, RAPD and the anterior segment.
- 2** If the retina cannot be visualised, arrange urgent ophthalmic review and B-scan ultrasonography to assess detachment.
- 3** Treat the cause: PDR, a PVD-associated tear, RVO, trauma or another retinal lesion; decide observation versus vitreoretinal intervention.

Do not anchor on diabetes

Context	Important competing cause
Diabetes + surface new vessels	PDR-related haemorrhage
Acute flashes + high myopia	PVD-associated tear, with or without detachment
Recent trauma	Retinal break or other traumatic injury

Escalation

Persistent or recurrent haemorrhage, traction threatening the macula, or a retinal break/detachment may require vitrectomy or other retinal treatment. Anticoagulant use alone is not an adequate explanation for the bleed.

References AAO: PVD and Retinal Breaks PPP (2025) | NEI: Retinal detachment and urgent assessment | NICE NG242: Diabetic retinopathy